YEAR 11 PHYSICS, AQA GCSE, LESSON 7 OF 10

Atoms & Nuclear Radiation.

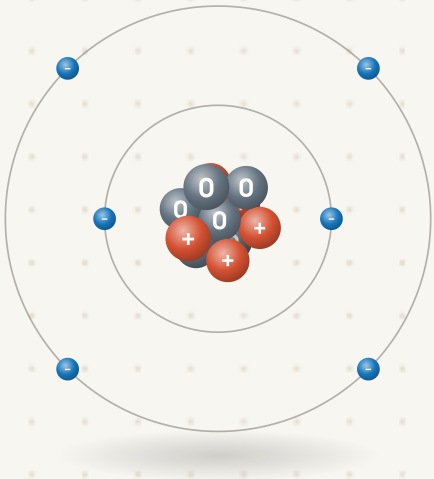
Lesson 07 Properties and uses of radiation

A smoke alarm uses alpha radiation, a paper mill uses beta and a hospital uses gamma. Why not use the same radiation for all three?

IN THIS LESSON YOU WILL

- compare the ionising power, range in air and penetration of alpha, beta and gamma
- explain why more ionising radiation is less penetrating
- identify radiation from absorber data
- choose a suitable source for a use and justify the choice

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lesson 7** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this lesson is highlighted)

01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

05 Radioactive decay

06 Nuclear equations

07 **Properties and uses of radiation** Ionising power, range and penetration; choosing the right source (slides 32–34)

08 Half-life and random decay

09 Half-life calculations

10 Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

07 Properties and uses of radiation.

AQA Physics 4.4.2.1; Combined Science 6.4.2.1

BIG QUESTION

A smoke alarm uses alpha radiation, a paper mill uses beta and a hospital uses gamma. Why not use the same radiation for all three?

BY THE END YOU CAN

- compare the ionising power, range in air and penetration of alpha, beta and gamma
- explain why more ionising radiation is less penetrating
- identify radiation from absorber data
- choose a suitable source for a use and justify the choice

ionising · ion · penetration · range · absorbed · smoke alarm · thickness gauge · tracer

SLIDES AND SIMULATIONS

Teaching deck, slides 32–34

Comparing alpha, beta and gamma, penetration, and choosing a source.

panphy.app/gcsephy/decks/atoms-and-radiation/#32

Nuclear Decay simulation

panphy.app/simulations/nuclear_decay.html

RETRIEVE

Do now

From earlier lessons. No notes!

1 Complete: $^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + ?$

$^{4}_{2}He$ (an alpha particle)

2 What happens to the atomic number in beta decay?

It increases by 1.

3 What is gamma radiation?

Electromagnetic radiation from the nucleus.

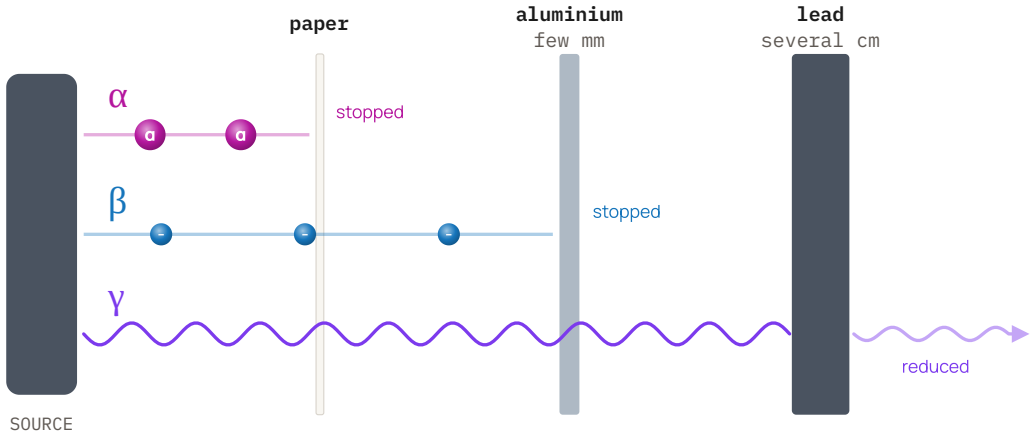
EXPLAIN

Ionising and penetrating

Nuclear radiation is ionising

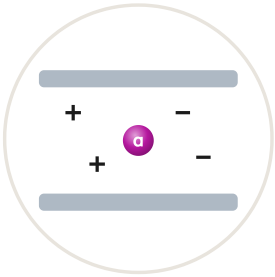
As nuclear radiation passes through a material, it knocks electrons off atoms, turning them into **ions** (Lesson 2). Each ionisation takes some of the radiation's energy. Alpha particles are large with a +2 charge, so they collide with many atoms and ionise strongly: they lose their energy fast and do not travel far.

	Alpha (α)	Beta (β)	Gamma (γ)
What is it?	helium nucleus	fast electron	EM wave
Charge	+2	−1	0
Ionising	strongly	moderately	weakly
Range in air	a few cm	about 1 m	very far
Stopped by	a sheet of paper, or skin	a few mm of aluminium	reduced by thick lead or concrete

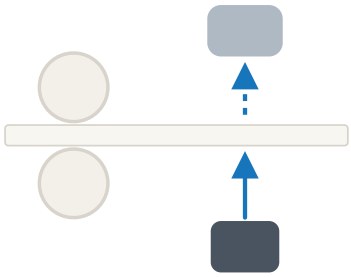


More ionising → less penetrating. Alpha loses its energy fastest, so it is stopped most easily. Gamma ionises weakly, so it travels furthest.

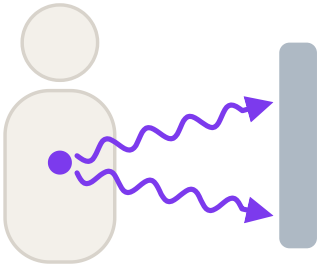
Choosing the right source



Alpha: smoke alarm
Alpha ionises the air, so a small current flows. Smoke absorbs the alpha particles, the current falls and the alarm sounds.



Beta: thickness gauge
Paper absorbs some beta. If the paper gets thicker, the count rate falls, and the rollers adjust.



Gamma: medical tracer TRIPLE ONLY
Gamma passes out of the body to a detector and is only weakly ionising.

To choose a source, think about **penetration** (must it pass through something, or be stopped?), **ionising power**, and **half-life**: how long it keeps giving out radiation (Lesson 8).

Identifying radiation from absorbers

A detector measures the count rate from a source as different absorbers are placed in between. The background count rate is 30 counts per minute.

WORKED EXAMPLE

Which radiation does this source emit?

Absorber	Count rate / counts per minute
none	900
paper	460
5 mm aluminium	455
2 cm lead	90

PAPER	900 → 460: a big fall, so alpha is present
ALUMINIUM	460 → 455: no real change, so no beta
LEAD	455 → 90: falls, but still above background (30), so gamma
ANSWER	**alpha and gamma**

YOUR TURN

Which radiation does this source emit?

Absorber	Count rate / counts per minute
none	520
paper	515
5 mm aluminium	32
2 cm lead	31

PAPER	no real change: no alpha
ALUMINIUM	falls to background: beta
LEAD	no change: no gamma
ANSWER	beta only

PRACTISE

Try it

1 SUPPORTED Complete the table.

	Alpha	Beta	Gamma
Ionising power	strong	moderate	weak
Stopped by	paper / skin	a few mm of aluminium	(reduced by) thick lead
Range in air	a few cm	about 1 m	very far

2 Which radiation is the most strongly ionising? Tick **one** box.

- [x] alpha
- [ ] beta
- [ ] gamma

3 Draw a line from each use to the radiation it needs.

1 smoke alarm	→ B	**A** gamma
2 paper thickness gauge	→ C	**B** alpha
3 medical tracer	→ A	**C** beta

4 Explain why a beta source is used to monitor the thickness of paper, and not alpha or gamma.

Alpha would be stopped completely by the paper, so the count rate would not change with thickness. Gamma would pass straight through almost unaffected. Beta is partly absorbed, so the count rate changes when the thickness changes.

5 ON YOUR OWN The background count rate is 20 counts per minute. Which radiation does this source emit? Explain using the data.

Absorber	none	paper	5 mm aluminium	2 cm lead
Count rate / counts per minute	640	380	25	21

Paper: 640 → 380, so alpha is present. Aluminium: 380 → 25, about background, so beta is present. Lead: no further change, so no gamma. The source emits alpha and beta.

6 Correct Mia's statement.

Mia: Gamma goes through everything, and nothing can stop it.

Gamma is very penetrating, but thick lead or concrete absorbs most of it: it is reduced, even if not stopped completely.

7 CHALLENGE Smoke alarms use americium-241, which has a half-life of 432 years. Suggest why a long half-life is useful here.

The source keeps giving out alpha radiation at an almost steady rate for many years, so the alarm works for its whole lifetime without replacing the source.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: Link each property to the reason: ionising → loses energy quickly → short range.

8 Explain why alpha radiation is more strongly ionising but less penetrating than gamma radiation. [3 marks]

Alpha particles are large with a +2 charge, so they collide with / ionise many atoms (1). Each ionisation transfers energy, so alpha loses its energy quickly (1) and is stopped by a short distance (paper or a few cm of air). Gamma has no charge, ionises weakly, so it loses energy slowly and travels much further (1).

9 A factory uses a beta source and a detector to control the thickness of aluminium foil. Explain how the count rate shows that the foil has become too thick. [3 marks]

Thicker foil absorbs more beta radiation (1), so fewer beta particles reach the detector (1) and the count rate falls below its normal value (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Nuclear radiation ionises atoms by knocking off electrons.
- Alpha: strongly ionising, a few cm in air, stopped by paper.
- Beta: moderately ionising, about 1 m in air, stopped by a few mm of aluminium.
- Gamma: weakly ionising, very far in air, reduced by thick lead or concrete.
- More ionising → less penetrating. Choose a source by penetration, ionising power and half-life.

COVER THE PAGE AND ANSWER

1 What does ionising mean?

2 What stops beta radiation?

3 Which radiation is used in a smoke alarm?

4 Why is gamma used as a medical tracer?

I CAN…

- compare the properties of alpha, beta and gamma
- explain the link between ionising and penetration
- identify radiation from absorber data
- choose and justify a source for a use

REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

Nuclear equations

Lesson 6

Emitted	Symbol	Mass no.	Atomic no.
alpha	$^{4}_{2}\mathrm{He}$	− 4	− 2
beta	$^{0}_{-1}\mathrm{e}$	no change	+ 1
gamma	γ	no change	no change
neutron	$^{1}_{0}\mathrm{n}$	− 1	no change

Top numbers balance; bottom numbers balance.

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

Properties of radiation

Lesson 7

	Alpha	Beta	Gamma
Ionising	strongly	moderately	weakly
Range in air	a few cm	about 1 m	very far
Stopped by	paper, skin	a few mm of aluminium	reduced by thick lead or concrete
Typical use	smoke alarm	thickness gauge	medical tracer

More ionising → less penetrating.

REFERENCE

Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Ionising: Able to knock electrons off atoms, turning them into ions.

Penetration: How far radiation can travel through materials before it is absorbed.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. $^{4}_{2}\mathrm{He}$ (an alpha particle)
2. It increases by 1.
3. Electromagnetic radiation from the nucleus.

COVER THE PAGE

1. Knocking electrons off atoms, making ions.
2. A few mm of aluminium.
3. Alpha.
4. It passes out of the body to a detector and is only weakly ionising.